



香港中文大學(深圳)  
The Chinese University of Hong Kong, Shenzhen

# PHY1001 Mechanics

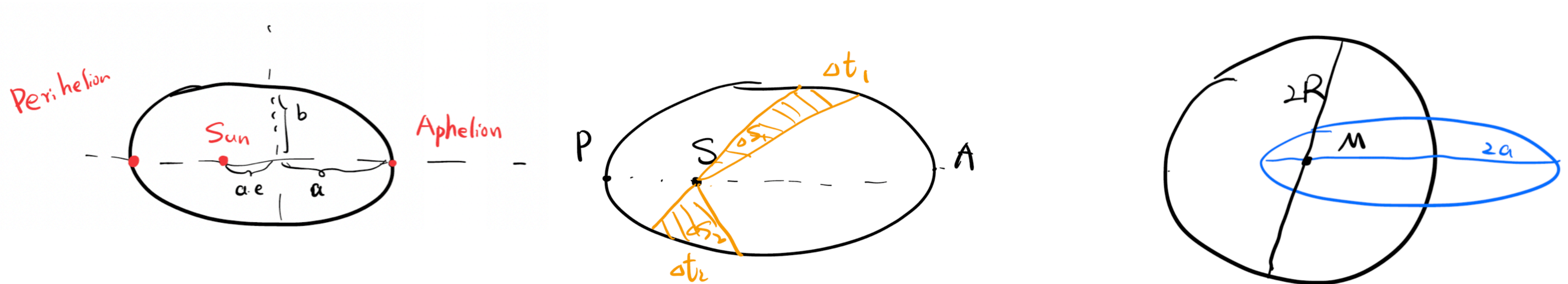
## Lecture 20: Gravitation, Fluid

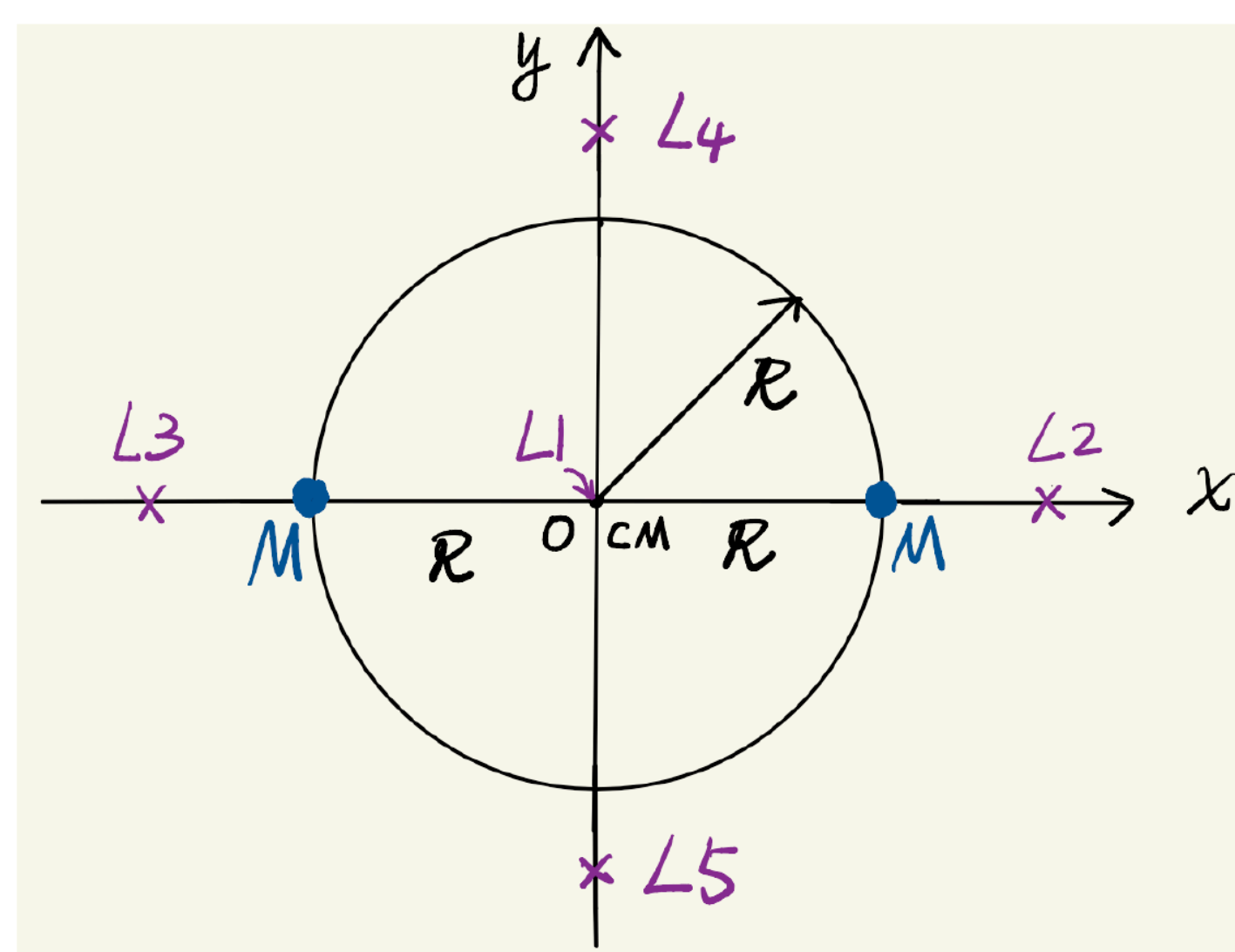
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# Recap: Kepler's Laws

- Kepler's Laws
  - First law: Orbits are ellipses with the Sun at the focus
  - Second Law: equal time, equal area
  - Third Law: Period proportional to  $3/2$  of semi-major axis length

↳ major axis same if





(e) Following part (d), will the locations of  $L_4$  and  $L_5$  change as compared to the result in part (c)? (Yes/No) **(1 pt)**

Hint: You do not have to give a reason. A Yes/No answer is sufficient.

$L_4 (0, \sqrt{3}R)$  two-star with same mass  $M$

$$\vec{F} = \vec{F}_1 + \vec{F}_2$$

$$\vec{F}_1 = G \frac{M_1 m}{(2R)^2} \hat{r}_1$$

$$= G \frac{M_1 m}{(2R)^3} \hat{r}_1$$

$$\vec{F} = G \frac{m}{(2R)^3} (M_1 \vec{r}_1 + M_2 \vec{r}_2)$$

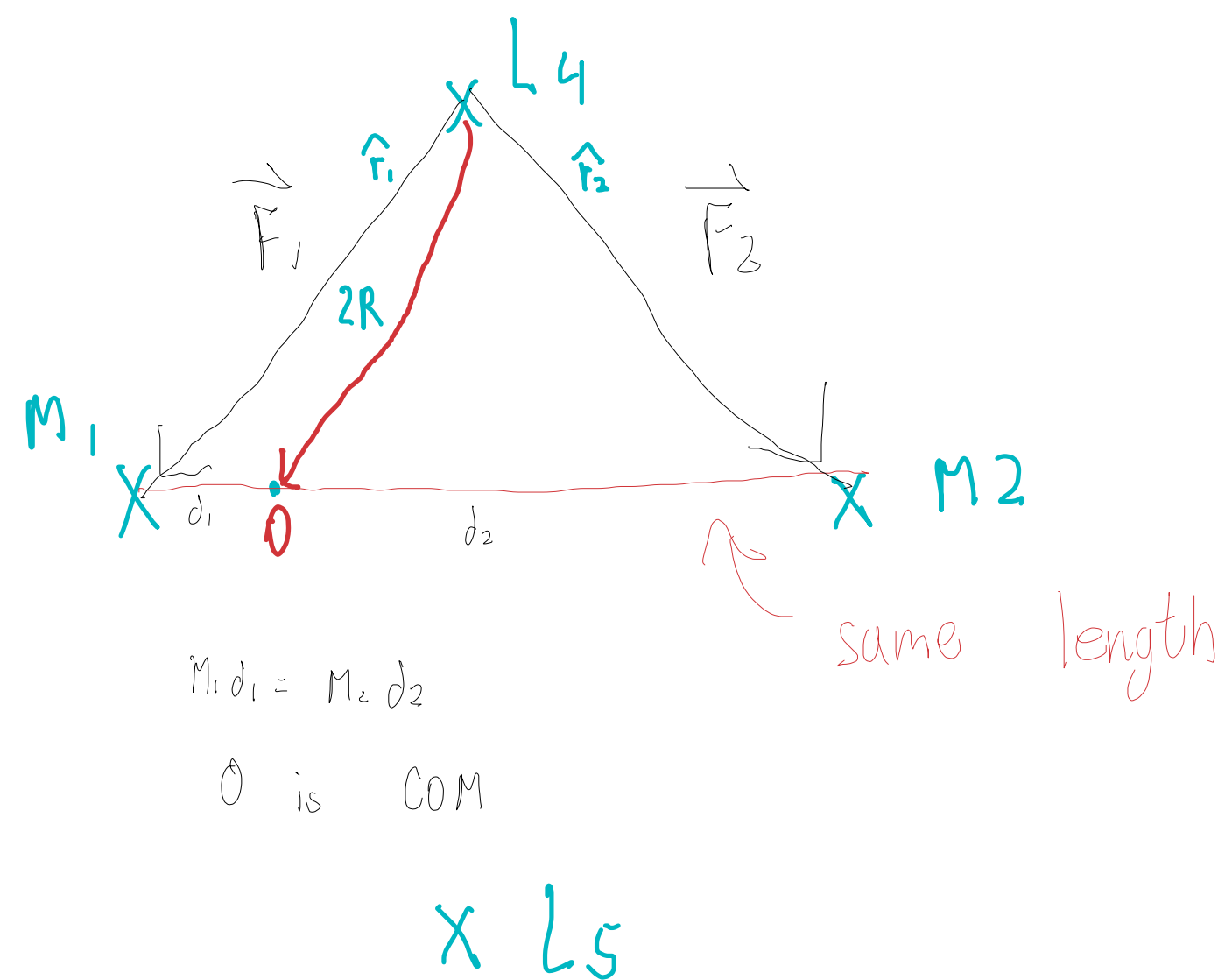
$$= G \frac{m}{(2R)^3} (M_1 + M_2) \vec{r}_{cm}$$

$$\vec{F}_2 = G \frac{M_2 m}{(2R)^3} \vec{r}_2$$

$$COM \quad \vec{r}_{cm} = \frac{M_1 \vec{r}_1 + M_2 \vec{r}_2}{M_1 + M_2}$$

$$\omega = \sqrt{\frac{G(M_1 + M_2)}{8R^3}}$$

$$\vec{F}_{centri} = m \omega^2 \vec{r} = m \frac{G(M_1 + M_2)}{8R^3} \cdot \vec{r}_{cm}$$

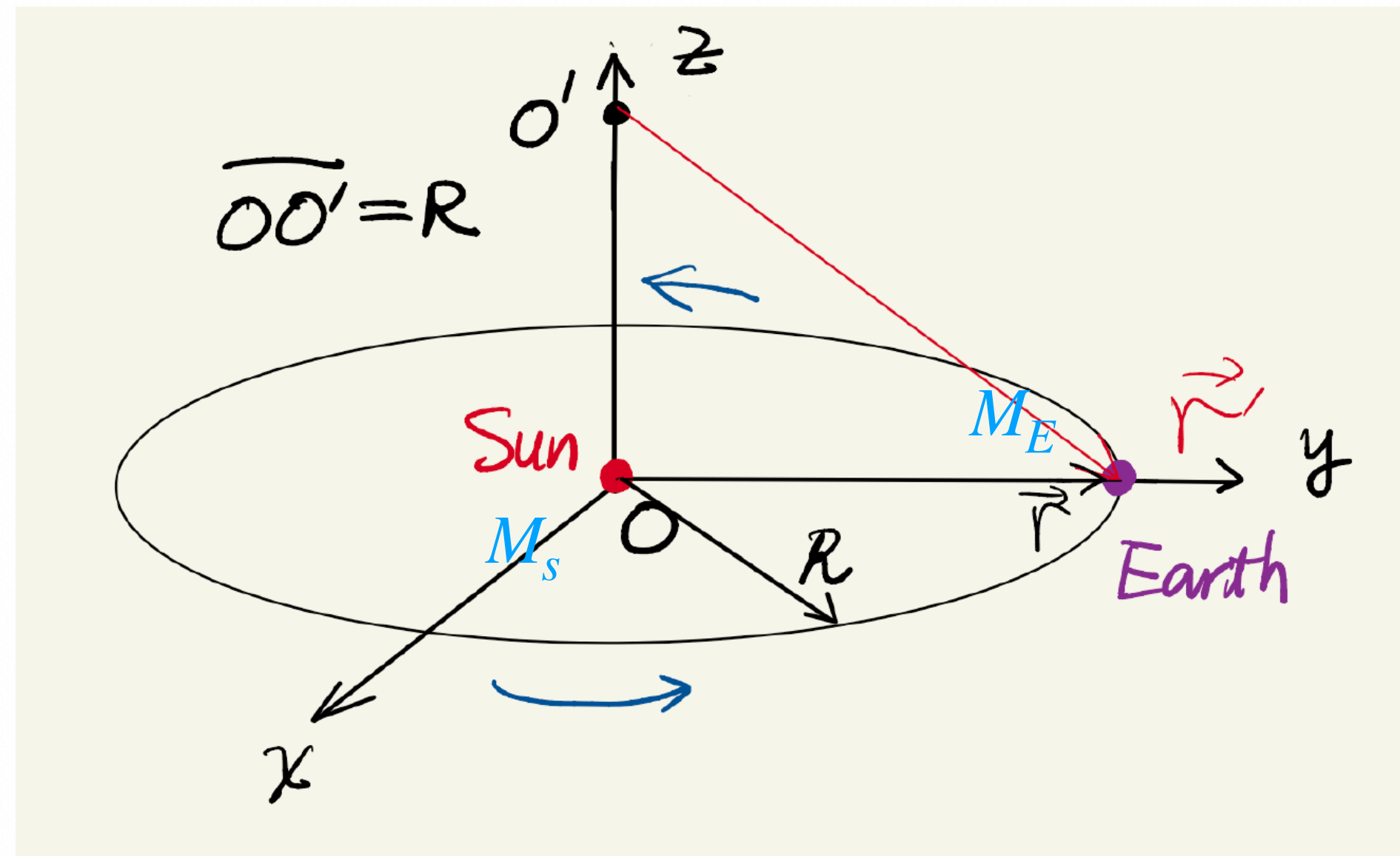




# Q10, Final 2024

## 10. Orbital Angular Momentum of the Earth

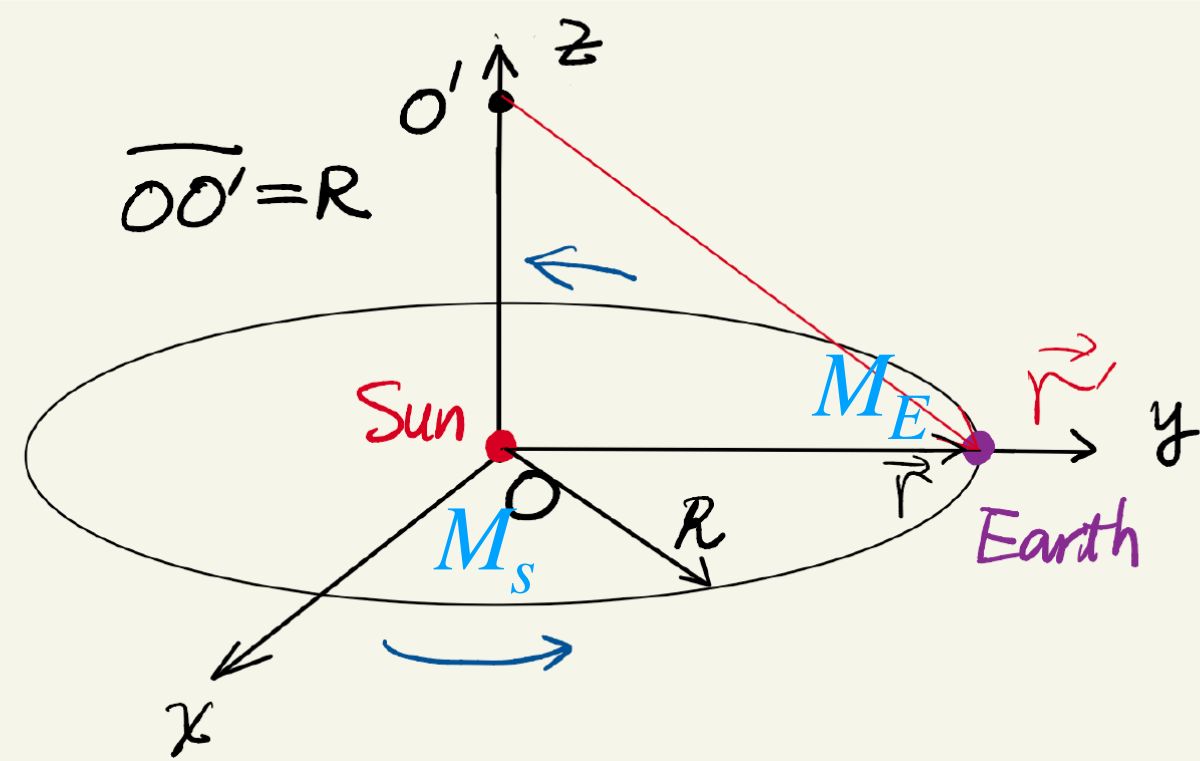
(10 pts)



Consider the Sun-Earth system with masses  $M_S$  and  $M_E$ , assuming  $M_S \gg M_E$  and neglecting other planets and moons. Treat the Earth as a point particle and its orbit as a circular orbit with a radius of  $R$ . In the diagram above, the Sun is located at point  $O$ , while point  $O'$  is directly above  $O$  at a distance of  $R$ . At one certain moment, the earth is on the  $y$  axis, and moving toward the  $-x$  direction. (You may refer to this position when you solve for part (c) and part (d)).

(a) Find the period  $T_0$  it takes for the earth to finish one turn (circle).

(2 pts)

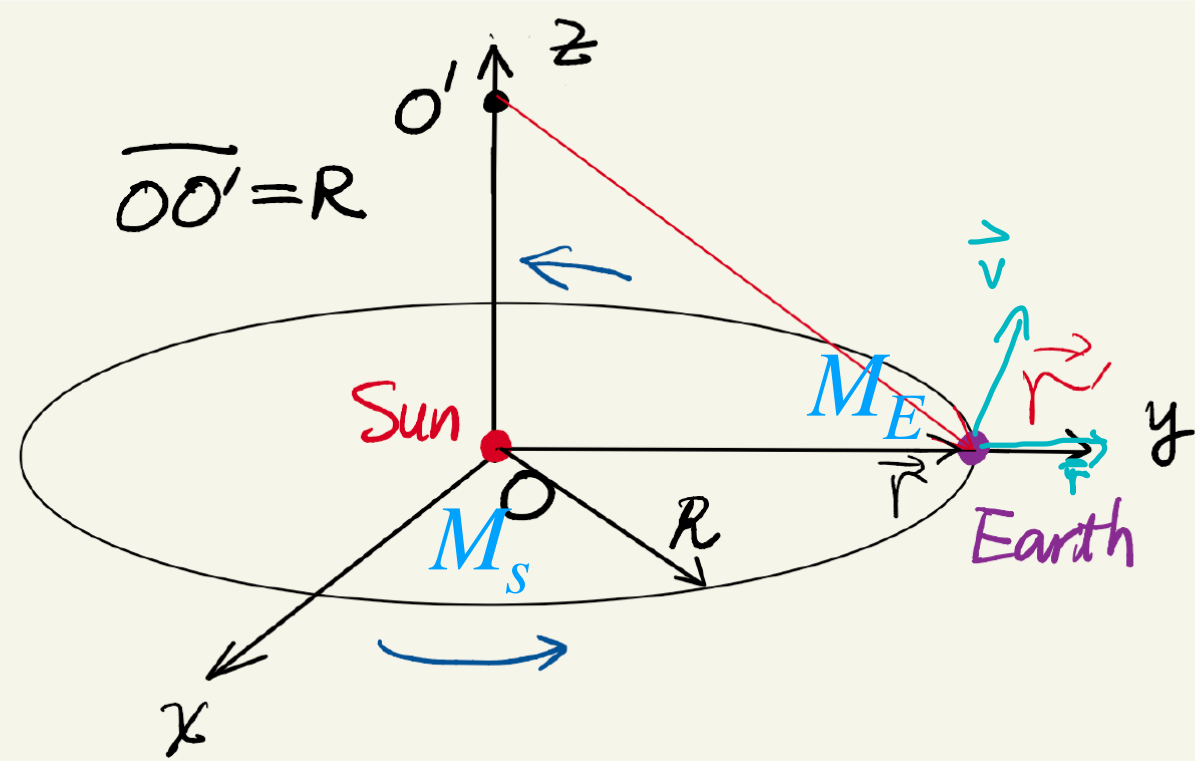


(a) Find the period  $T_0$  it takes for the earth to finish one turn (circle).

**(2 pts)**

$$G \frac{M_s M_E}{R^2} = M_E \left( \frac{2\pi}{T_0} \right)^2 R$$

$$T_0 = 2\pi \sqrt{\frac{R^3}{G M_s}}$$



- (b) Choose  $O$  as the reference point and compute the orbital angular momentum  $\vec{L}_O$  of the earth. Hint: Provide both the magnitude and direction according to  $\vec{L}_O = M_E \vec{r} \times \vec{v}$ . **(2 pts)**

Direction :  $\uparrow z$  ( $\hat{k}$ )

$$|\vec{L}| = M_E |\vec{v}| \cdot |\vec{r}| \cdot \sin \theta$$

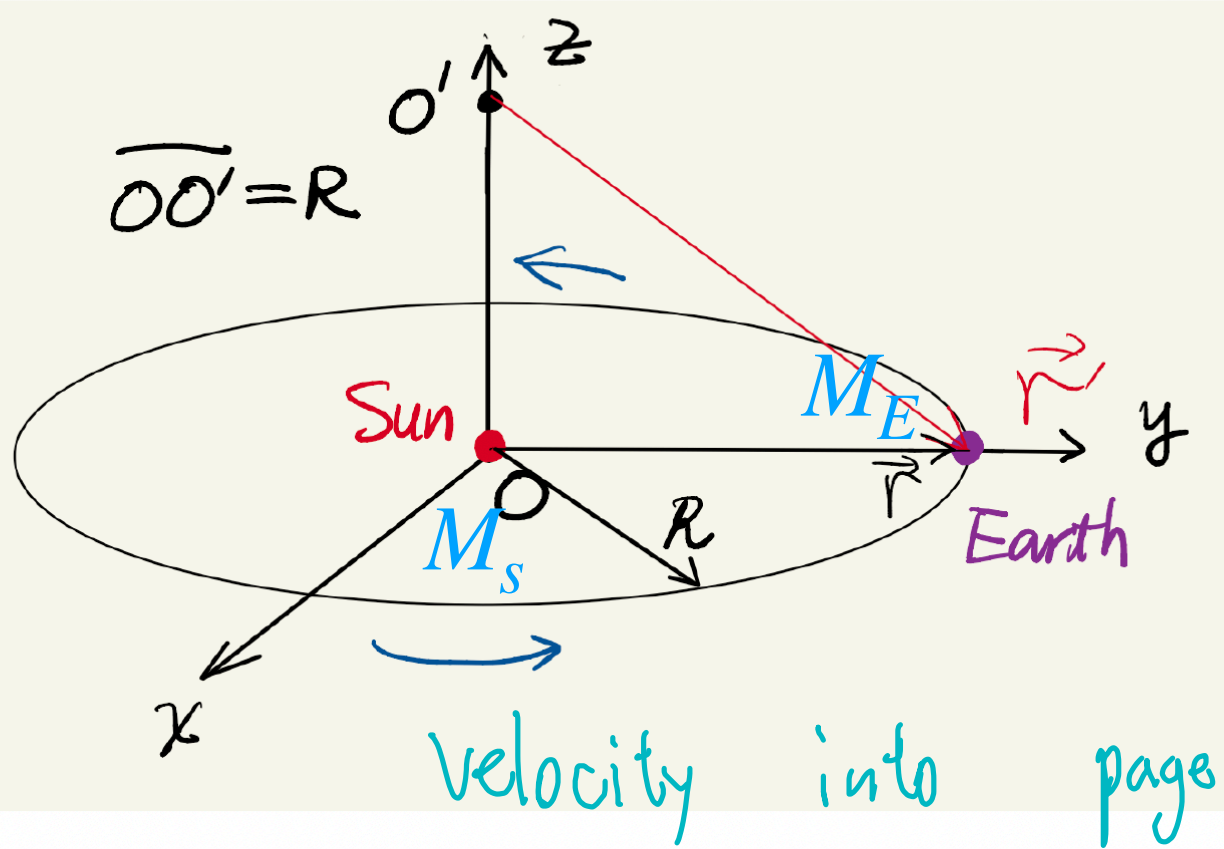
$$= M_E \cdot |\vec{v}| \cdot R$$

$$= M_E \cdot \frac{2\pi R}{2\pi \sqrt{\frac{R^3}{GM_S}}} \cdot R$$

$$= M_E \cdot \sqrt{GM_S R}$$



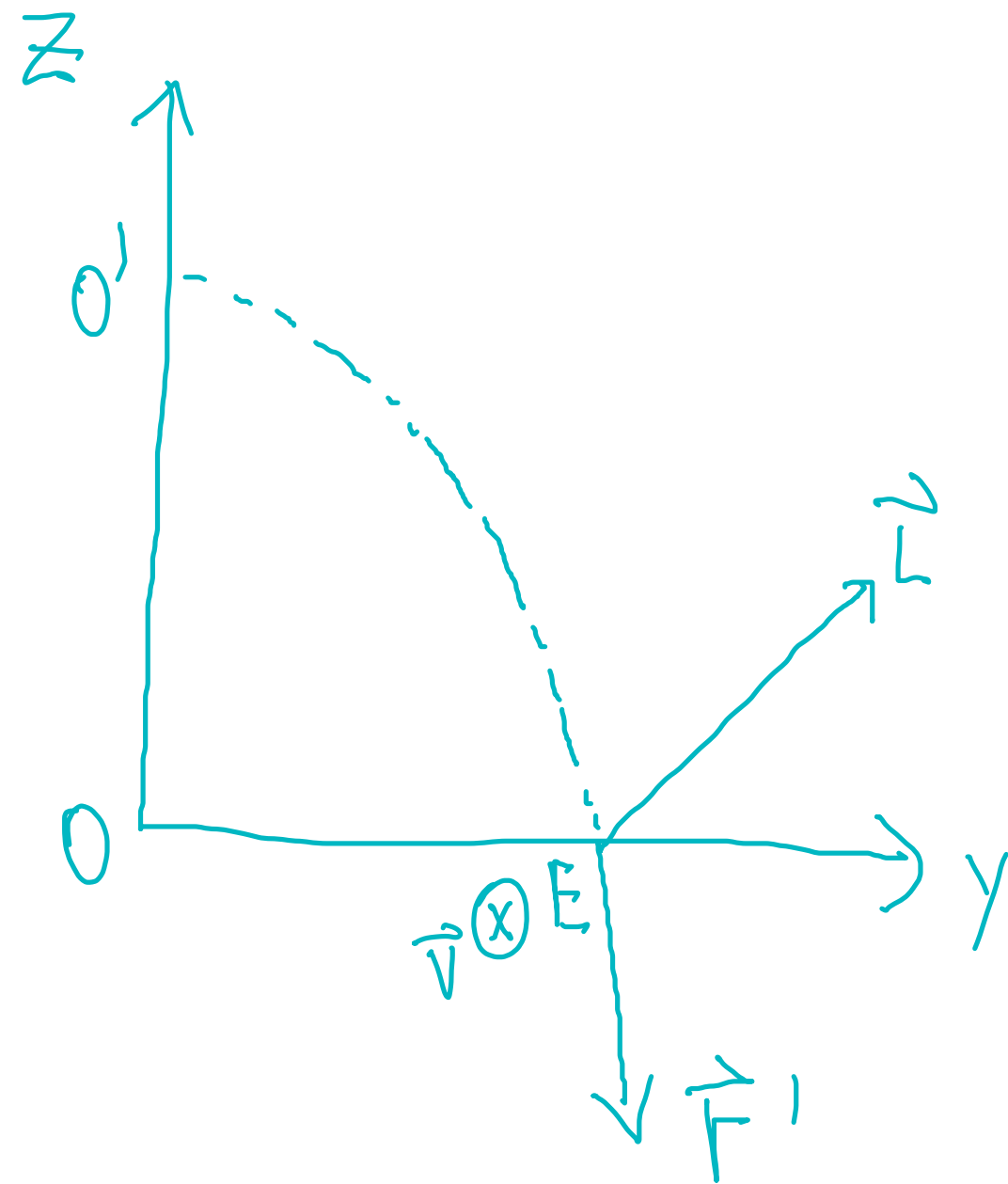
- (c) Now choose  $O'$  as the reference point, and compute the orbital angular momentum  $\vec{L}$  of the earth. Hint:  $\vec{L}$  is different from  $\vec{L}_0$ . Give the magnitude and draw the direction. **(2 pts)**



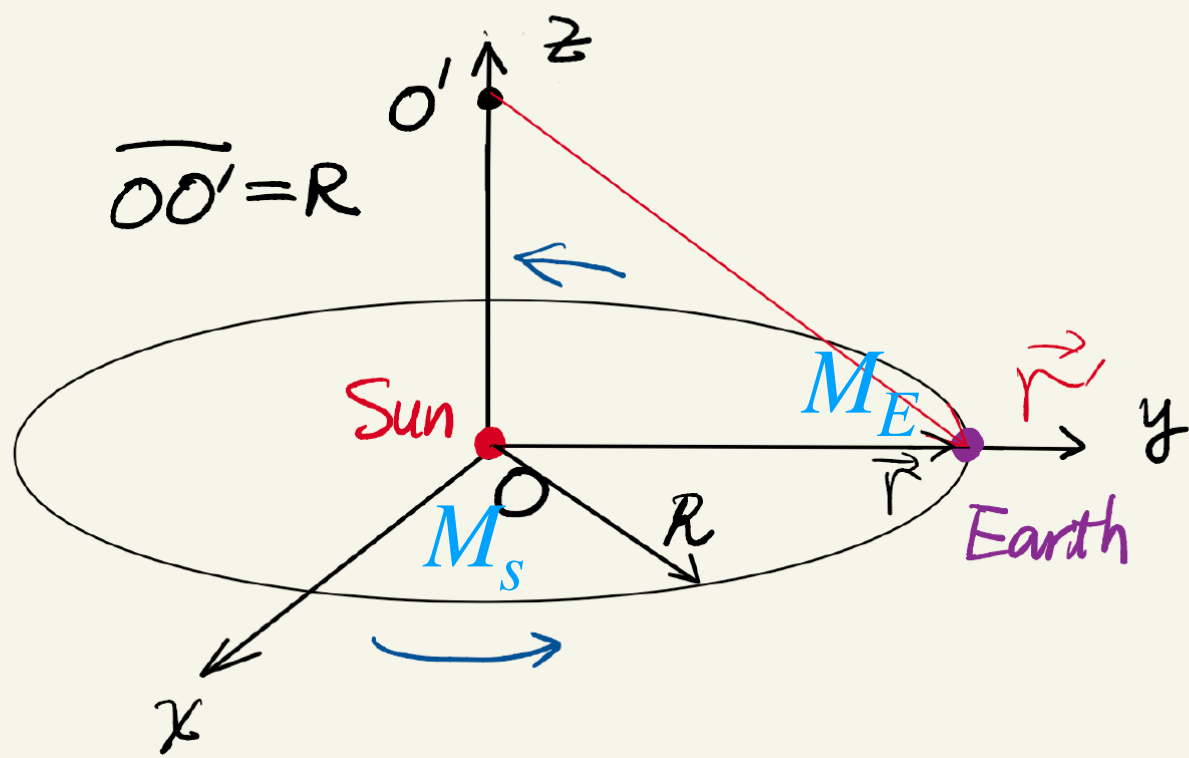
$$|\vec{L}| = M_E \cdot |\vec{v}| \cdot |\vec{r}|$$

$$= M_E \cdot \sqrt{\frac{GM_s}{R}} \cdot \sqrt{2} R$$

$$= M_E \sqrt{2GM_s R}$$



- (d) Compute the torque  $\vec{\tau} \equiv \vec{r}' \times \vec{F}$  about the reference point  $O'$  generated by the gravitational pull (force) of the Sun on the Earth ( $\vec{F}$ ). **(2 pts)**  
 Hint: Give the magnitude and direction!

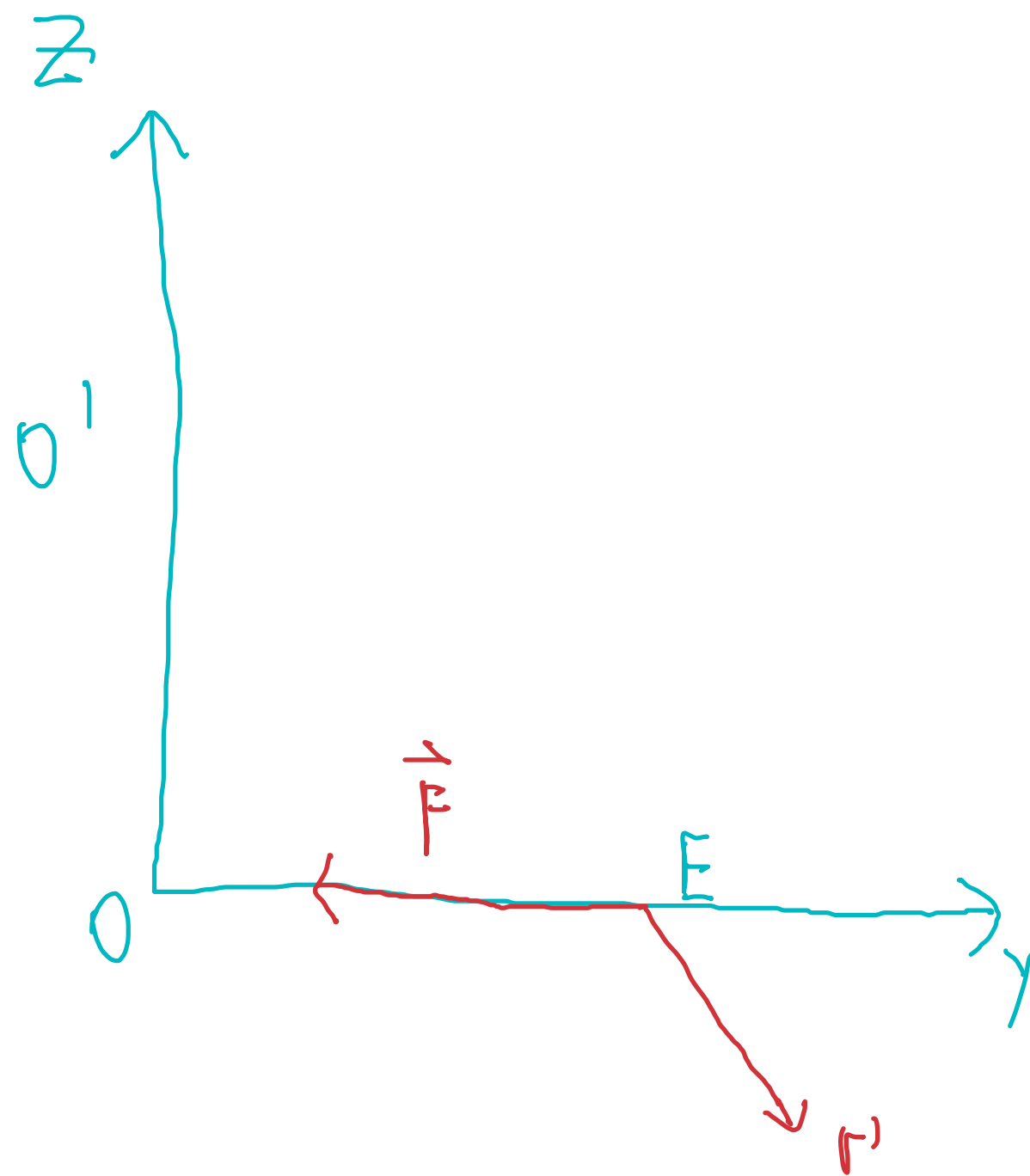


direction : - x axis

$$|\vec{\tau}| = |\vec{r}'| \cdot |\vec{F}| \cdot \sin \frac{3}{4} \pi$$

$$= \sqrt{2} R \cdot G \frac{M_E M_S}{R^2} \cdot \frac{\sqrt{2}}{2}$$

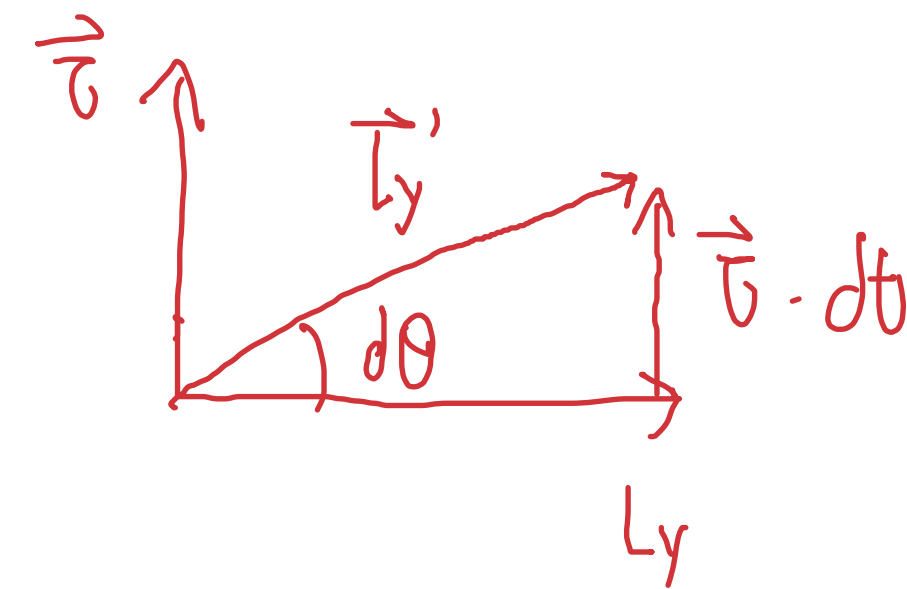
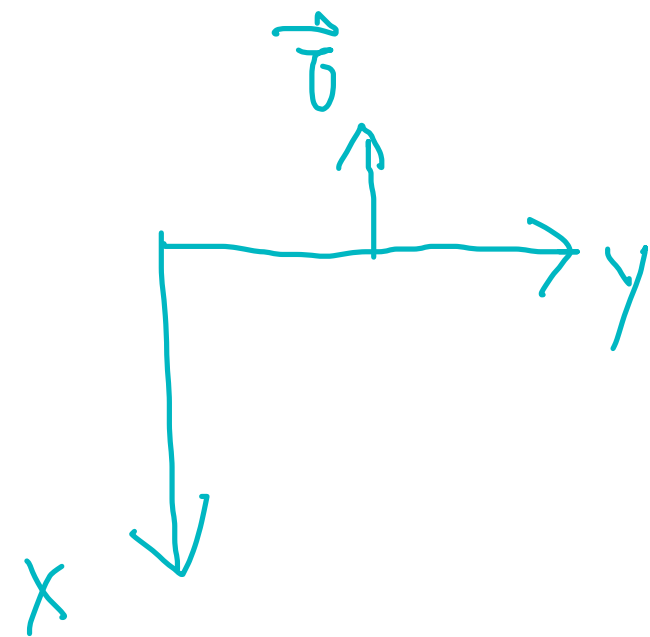
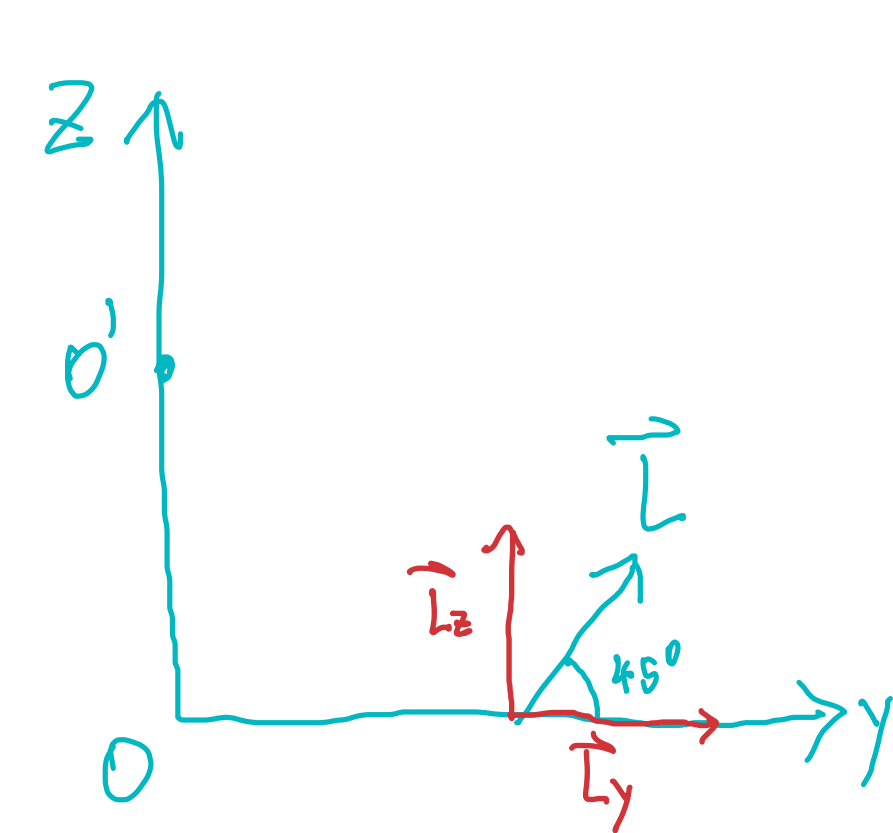
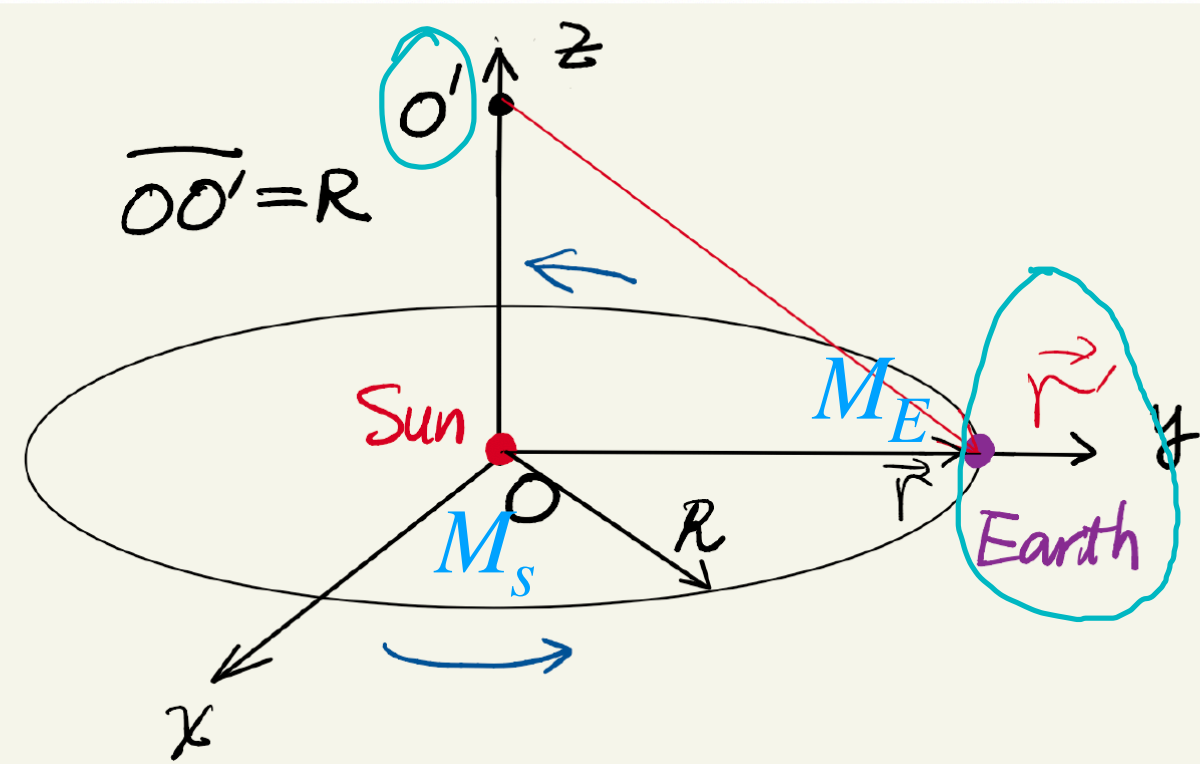
$$= G \frac{M_E M_S}{R}$$





- (e) Choose  $O'$  as the reference point, then use Newton's second law for rotation  $\vec{\tau} = \frac{d\vec{L}}{dt}$  to explain the change of the orbital angular momentum  $L$ . Compute the angular frequency  $\Omega$  of this "rotation" of the orbital angular momentum. **(2 pts)**

Use the idea of "precession"



$$\frac{d\theta}{dt} = \Omega = \frac{|\vec{\tau}| dt}{|\vec{L}_y|} = \frac{|\vec{\tau}|}{|\vec{L}_y|}$$

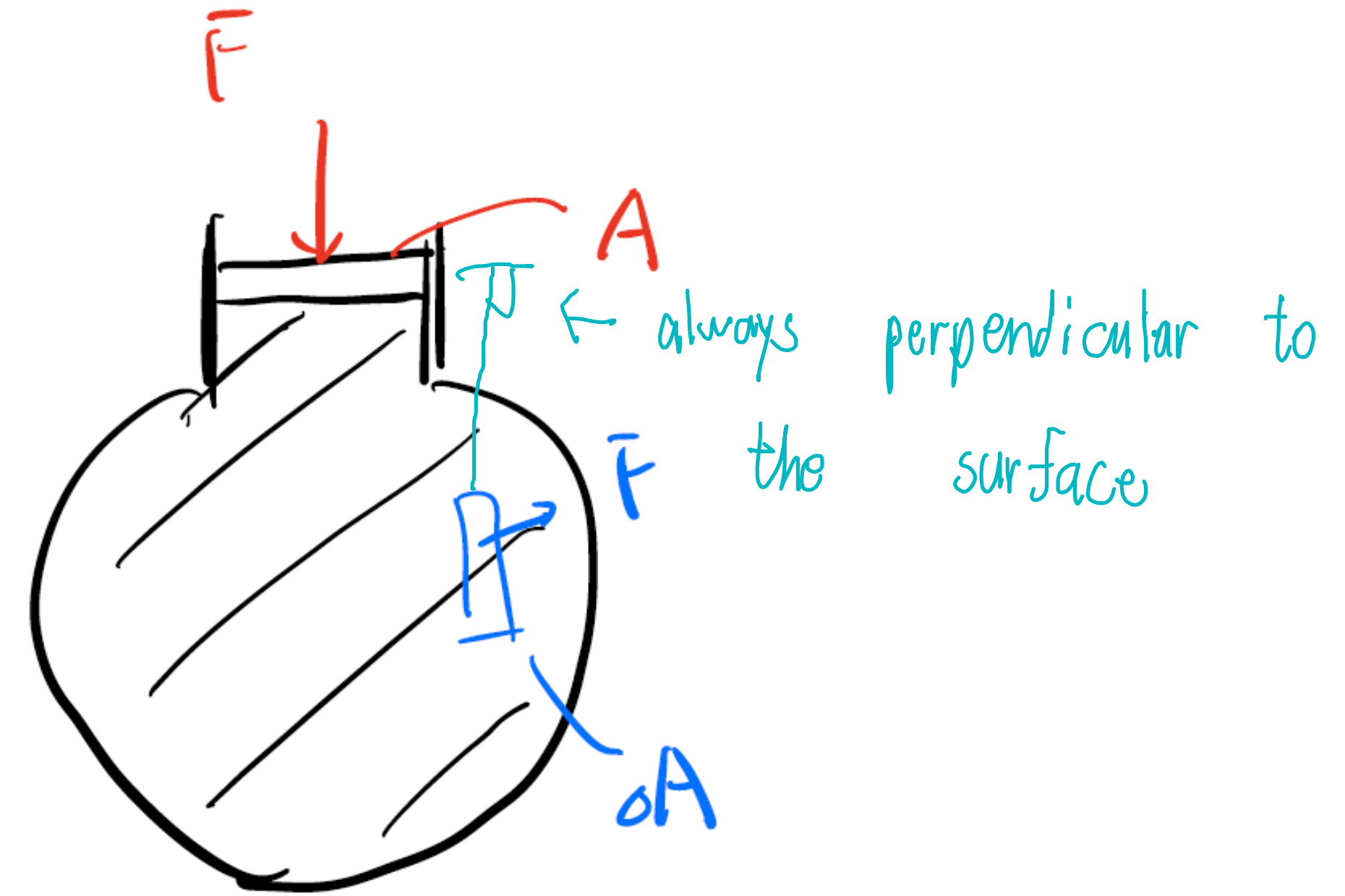
$$= \frac{G \frac{M_E M_S}{R}}{M_E \sqrt{G M_S R}} = \sqrt{\frac{G M_S}{R^3}}$$

$$\omega_{Earth} = \frac{2\pi}{T_0} = \frac{G M_S}{R^3}$$

# Fluid

- Pressure  $P = \frac{F}{A}$ 
  - Unit:  $\text{N/m}^2$ , Pa

Static fluid



# Pascal's Principle

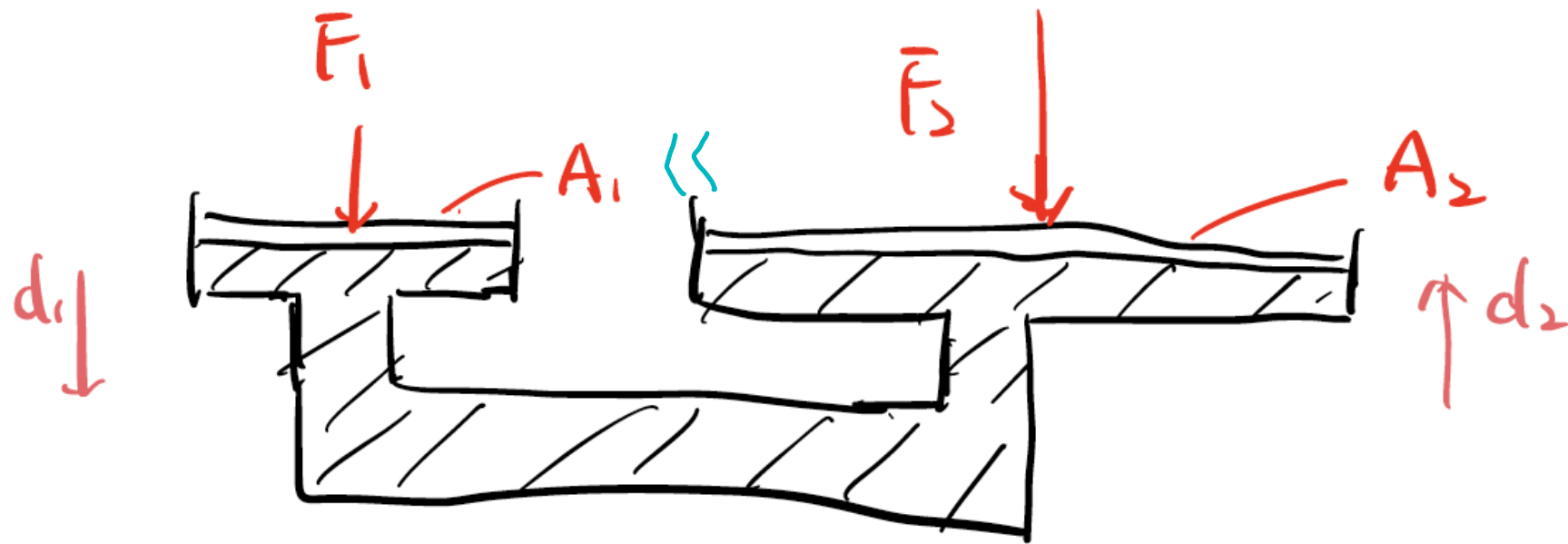
- Pascal's Principle: Pressure applied to an enclosed fluid is transmitted undiminished to every portion of the fluid.
  - Application: Hydraulic jack
- Incompressibility/continuity:  $A_1 d_1 = A_2 d_2$

$$F_1 = P A_1$$

$$F_2 = P A_2$$

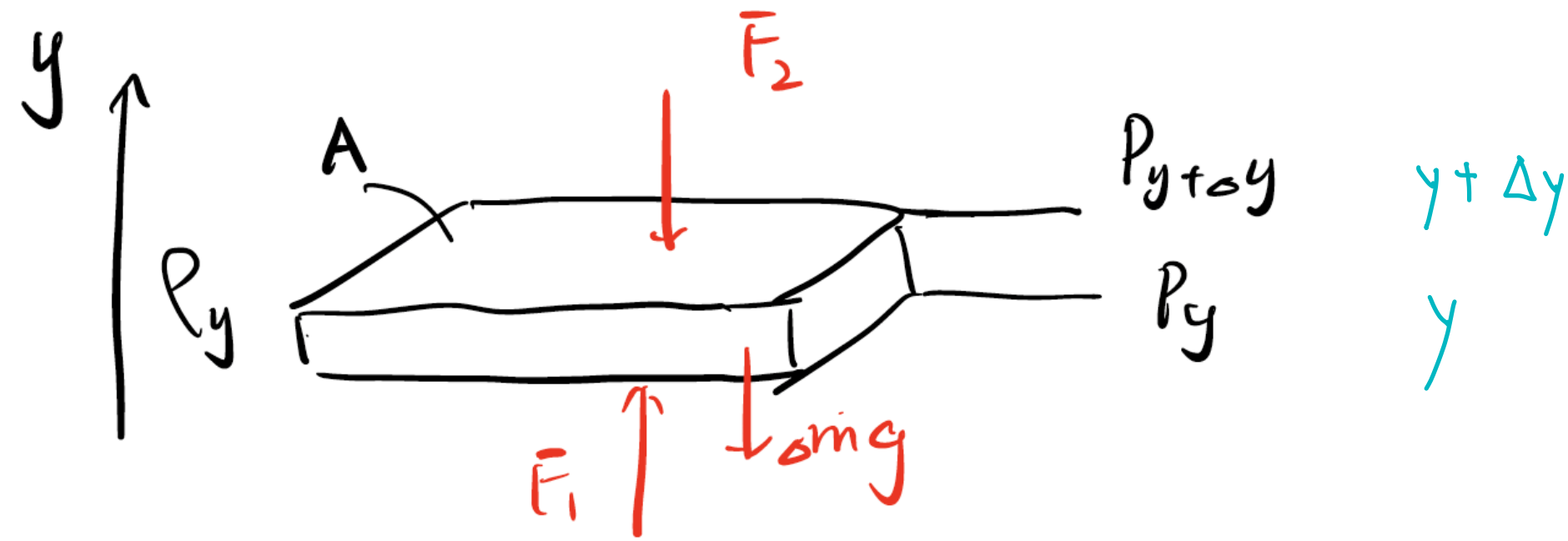
$$\frac{F_2}{F_1} = \frac{A_2}{A_1} = 100$$

$$\frac{d_1}{d_2} = \frac{A_2}{A_1} = 100$$





# Hydrostatic Pressure



$$F_1 = F_2 + \Delta mg$$

$$P_y \cdot A = P_{y+\Delta y} \cdot A + P_y \cdot A \cdot \Delta y \cdot g$$

$$-P_y g = \lim_{\Delta y \rightarrow 0} \frac{P_{y+\Delta y} - P_y}{\Delta y} = \frac{dP}{dy}$$

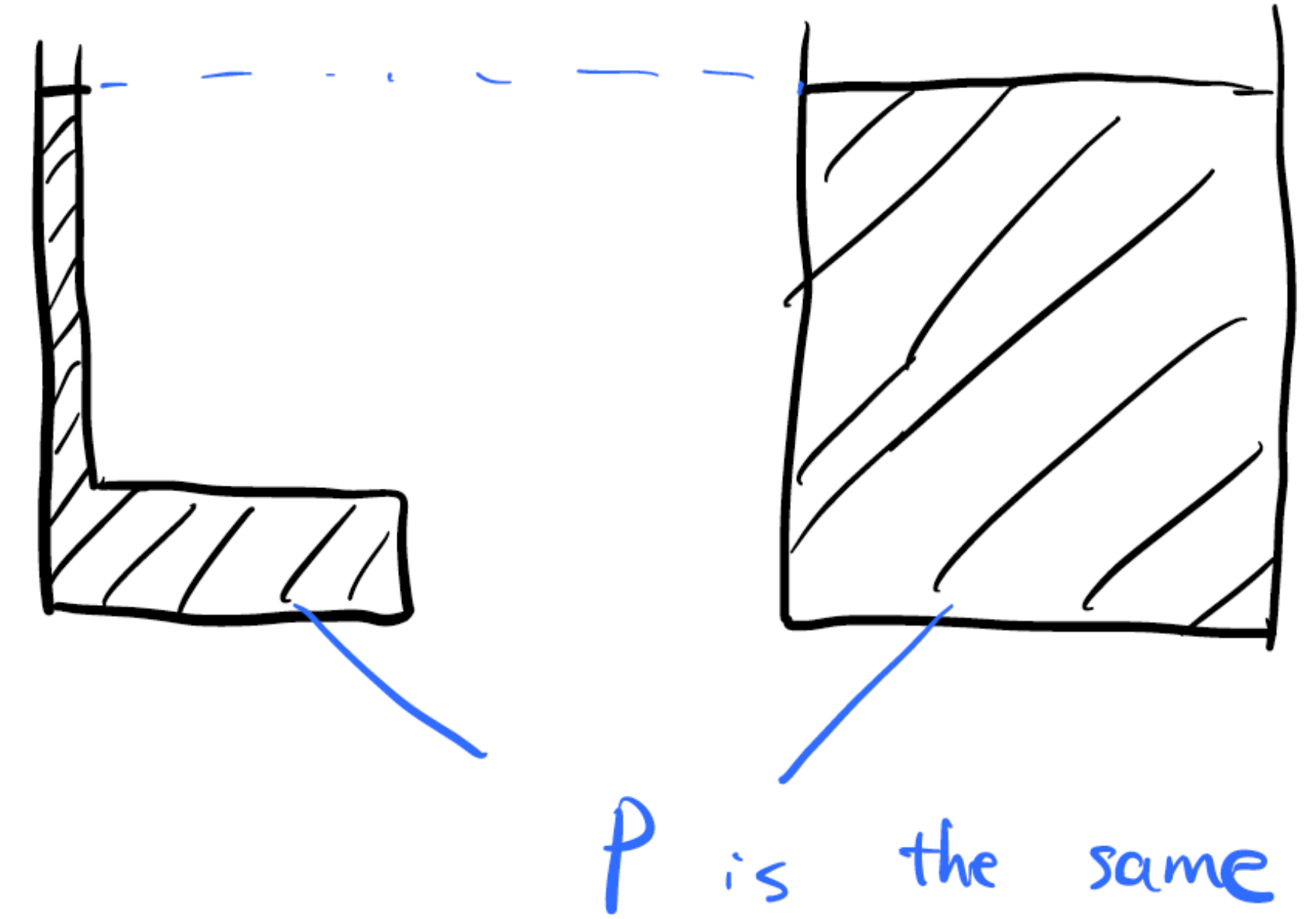
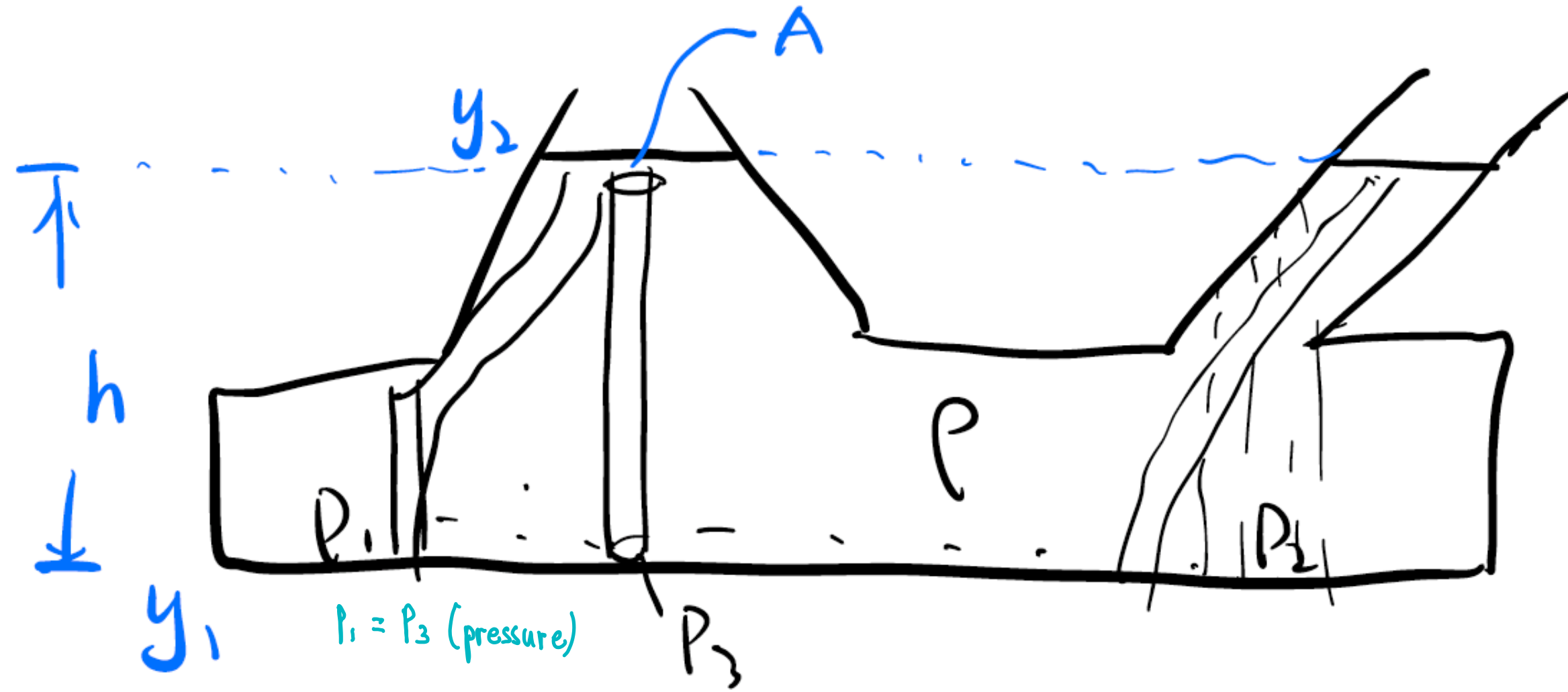
- Hydrostatic pressure:  $-\rho g = \frac{dP}{dy}$
- Assume liquid is incompressible:  $P_2 - P_1 = \rho g(y_1 - y_2)$

$$\int_{y_1}^{y_2} -\rho g = \int_{y_1}^{y_2} \frac{dP}{dy}$$

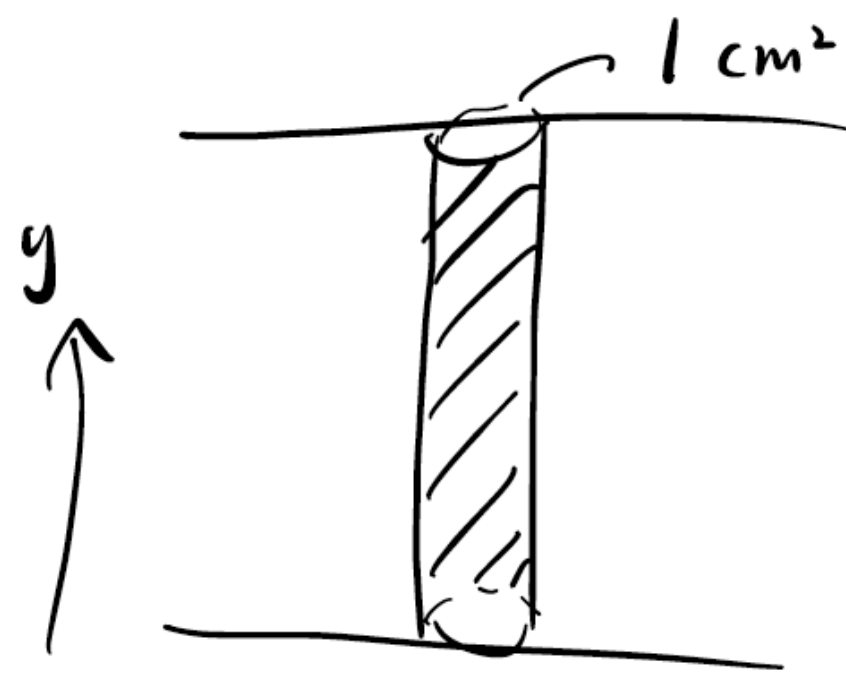
$$-\rho g (y_2 - y_1) = P_2 - P_1$$

$$\rho g (y_1 - y_2) = P_2 - P_1$$

# Consequences of Hydrostatic Pressure and Pascal's Principle



# Atmospheric/Barometric Pressure



$$\bar{F} = 10 \text{ N} / 1 \text{ cm}^2$$



# How to Measure Atmospheric Pressure?

